

Source-Known Identifiers: A Three-Tier Identity System for Distributed Applications

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ABSTRACT

Distributed applications use identifiers across database storage, trusted communication, and external access. These roles require different combinations of storage efficiency, chronological sortability, origin metadata embedding, zero-lookup verification, metadata confidentiality, and multi-century addressability. The identifier schemes compared in this paper, including Universally Unique Identifier (UUID) versions 4 and 7, do not individually provide all six properties. We present Source-Known Identifiers, a three-tier identity system that distributes these properties across representations of one entity identity, connected through deterministic transformations. A 64-bit Source-Known ID (SKID) provides a timestamp-ordered database key. A 128-bit Source-Known Entity ID (SKEID) adds entity type, epoch, and a keyed message authentication code (MAC) to the SKID, enabling integrity and origin metadata checks within a shared-key trust domain without record lookups. Secure SKEID encrypts this representation to conceal metadata while retaining linkability of repeated identifiers. Collision checks prevent generated ciphertext from being accepted as plaintext. Evaluation combines specification and threat analysis, implementation tests, and microbenchmarks of the C#/ .NET 10 reference implementation. Within sequence capacity on Apple M2 hardware, SKEID and Secure SKEID generation averaged 207.28 ± 3.10 ns and 224.62 ± 3.90 ns, respectively, compared with 361.94 ± 1.94 ns for UUID Version 4 and 384.11 ± 1.95 ns for UUID Version 7. $\pm$ values indicate 99.9% confidence interval margins. Under saturation, generation was slower than the UUID baselines. These results support the feasibility of deriving SKEIDs from 64-bit Source-Known IDs on demand.

1 INTRODUCTION

1.1 Problem Statement

Distributed applications use identifiers as database keys, internal references, and externally visible resource handles. These roles impose different requirements on storage size, ordering, integrity verification, and metadata exposure. This paper investigates how deterministic transformations can provide different representations of one entity identity across these boundaries.

An *entity* is a domain object with a distinct identity that persists across its lifecycle (Evans, 2003). Its *entity identifier* distinguishes it from other entities of the same type. While this terminology is established in Domain-Driven Design, the underlying requirement of uniquely and persistently identifying records across distributed systems is universal regardless of architectural style, design patterns, or definitions. Specifically, modern distributed applications require entity identifiers that serve multiple roles simultaneously such as database primary keys, inter-application correlation tokens, and externally visible resource handles. However, existing identifier schemes force a choice between conflicting properties.

Database sequences offer compact storage (4 or 8 bytes) and natural ordering but expose creation patterns when passed to external consumers. UUID Version 4 (Davis, Peabody & Leach, 2024) provides 122 bits of randomness for probabilistic uniqueness but sacrifices ordering. UUID Version 7 (Davis, Peabody & Leach, 2024) restores time-ordering with a 48-bit Unix timestamp but exposes creation patterns. Snowflake-style identifiers (Twitter Engineering, 2012) embed timestamp and worker topology in 64 bits but again expose creation patterns and lose multi-century addressability.

None of these schemes provides built-in verification of identifier integrity and authenticated origin metadata without a lookup. Exposing chronological order also reveals information that may need to remain confidential outside trusted environments.

A multi-tier identifier system can address these challenges by providing different identifier properties for different trust levels. The proposed system targets the following six properties across its tiers:

1. Storage efficiency
2. Chronological sortability
3. Origin metadata embedding
4. Zero-lookup verification
5. Confidentiality for external consumers
6. Multi-century addressability

This study asks under what deployment and cryptographic assumptions deterministic transformations can distribute the six target properties across representations of a single entity identity at different trust boundaries, and how generation costs across the multi-tier system compare with those of UUID versions 4 and 7. These UUID versions were selected as standardized baselines defined in RFC 9562, representing random and time-ordered identifiers, respectively.

1.2 Motivation

At the persistence tier, B-tree index performance favors sequential or time-ordered primary keys (Davis, Peabody & Leach, 2024, secs. 2.1, 6.11, and 6.13). However, exposing these identifiers through external APIs can make enumeration and exploitation of Insecure Direct Object Reference (IDOR) vulnerabilities easier (MITRE Corporation, 2026a). Exposure can also enable adversarial inference of generation velocity and record volume (the German Tank Problem (Ruggles & Brodie, 1947)). As a workaround, the dual-identifier pattern with an integer primary key alongside a random UUID alternate key compromises storage efficiency.

The full cost of such workarounds extends beyond the identifier columns themselves. For instance, a conventional schema using an auto-increment primary key (8 bytes), a UUID external identifier (16 bytes), and a `created_at` timestamp (8 bytes) requires 32 bytes of column data per record. In a fully indexed schema, each column demands a separate B-tree index, resulting in three indexes and their associated maintenance operations such as vacuum, reindex, and statistics collection. An identifier that embeds a timestamp and derives a UUID-compatible external representation deterministically at application runtime consolidates all three concerns into a single 8-byte primary key column with a single index, if the embedded generation timestamp satisfies the application's `created_at` requirement.

Under this fully indexed baseline, this represents a 75% reduction in raw column data across these three fields and a reduction from three single-column indexes to one. Total disk footprint savings depend on storage-engine tuple overhead (such as row headers and alignment padding) and page fill factors. Additional composite indexes involving these fields would amplify the savings further.

Where downstream applications require identifier authenticity verification (MITRE Corporation, 2026b), an embedded MAC enables verification of identifier integrity without a query or cache lookup for the corresponding record. This eliminates record-lookup I/O for that verification step.

Time-ordered identifiers (Davis, Peabody & Leach, 2024) can also support historical analysis when records from independent generators are combined. Their usefulness depends on timestamp precision, clock synchronization, and consistent epoch interpretation. When data from independent generators must be merged years or decades after creation, the identifier scheme must preserve consistent ordering beyond the operational lifespan of individual system components. This is especially critical for any system requiring long-term data retention and historical analysis.

These requirements motivate a multi-tier identifier system where a single entity identity is represented differently across trust boundaries.

1.3 Prior Work

Source-Known Identifiers build on established techniques for structuring and protecting identifiers. Snowflake packs timestamps, topology, and sequence counters into 64-bit identifiers (Twitter Engineering, 2012), while UUIDv8 permits application-defined fields within a standard 128-bit layout (Davis, Peabody & Leach, 2024, sec. 5.8). These provide precedents for SKID's topology fields and SKEID's metadata layout.

The cryptographic foundations are also established. MACs provide integrity verification, and authenticated encryption combines confidentiality with authenticity (Song et al., 2006; Harkins, 2008; McGrew, 2008).

Bellare and Rogaway's encode-then-encipher framework shows how checking embedded redundancy after deciphering can provide authenticity under a strong pseudorandom-permutation assumption (Bellare & Rogaway, 2000, Theorem 2). SKEID incorporates an embedded MAC, and Secure SKEID follows this principle by encrypting the structured representation and verifying it after decryption.

For external use, Khuong (2022) describes using AES to convert a structured 128-bit key into an opaque identifier of the same size. The article also discusses cycle walking and a specialized Feistel construction for producing encrypted identifiers with fixed format bits, including those required by UUIDv8. As discussed in Future Work, preserving UUID version and variant bits may be unnecessary where the complete application path accepts opaque 128-bit identifiers.

IDMask combines reversible identifier encryption with forgery checks (Favre, 2023). Its default deterministic 64-bit mode encrypts the identifier and 64 bits of validation redundancy in one AES block, then adds a version byte to produce a 17-byte output. Its default deterministic 128-bit mode adds an 8-byte MAC and a version byte to the encrypted identifier, producing 25 bytes. The MAC in the 128-bit mode remains visible outside the ciphertext. In both modes, the version byte is XOR-obfuscated with the first ciphertext byte and can be recovered without the key. Both outputs exceed 128 bits. IDMask is excluded from the comparison tables because it protects application-supplied identifiers, whereas the tables focus on identifier-generation schemes.

Source-Known Identifiers bring these techniques together across three representations of one identity, combining compact storage, authenticated entity-type and origin metadata, and encryption for external use. Conversion from SKEID to Secure SKEID preserves the 128-bit size. Collision-guard and backward-verification mechanisms distinguish the plaintext and encrypted representations.

1.4 Contributions

The contributions concern the three-tier architecture and its generation and parsing mechanisms. AES-256 and BLAKE3 are established primitives. The design combines them with embedded metadata, a collision guard, and backward verification while preserving the 128-bit size when converting SKEID to Secure SKEID.

1. **Design of a three-tier identity system** that satisfies all six desired identifier properties across its three tiers, aligned with trust boundaries (database, trusted internal, and external).
2. **A defense-in-depth security architecture** combining AES-256 encryption, BLAKE3 keyed MAC, marker byte detection, epoch validation, and optional entity type and application ID matching.
3. **A collision guard for plaintext-to-encrypted identifier conversion** that checks candidate ciphertext against plaintext acceptance criteria and, on collision, recomputes the MAC and re-encrypts successive variants. Successfully generated ciphertext cannot be accepted as plaintext.
4. **Cryptographic backward verification** that enables the parser to reconstruct and verify every preceding collision under the authenticated key pair before accepting a non-default variant. This mechanism verifies that every preceding variant would have triggered the collision guard, without trusting that an input was produced by the generator.
5. **Zero-expansion plaintext-to-encrypted identifier conversion** that combines embedded metadata, a truncated MAC, and collision-guard variants within the same 128-bit representation using AES's length-preserving block permutation. By comparison, AES-SIV and the AES-GCM profiles in RFC 5116 each add 16 bytes to the plaintext length, excluding any separately transmitted nonce (Harkins, 2008, sec. 2.6; McGrew, 2008, sec. 5.1 and 5.2).
6. **An epoch addressing system** embedded in the plaintext SKEID's UUIDv8-compatible layout that spans approximately 17,421 years of total coverage through 256 configurable epochs of 2^{31} seconds each (approximately 68 years per epoch).
7. **A reference implementation** in C#.NET 10 with open-source code, integration and unit test coverage, and BenchmarkDotNet performance data demonstrating lower mean generation times than standard UUID generation within sequence capacity, with separate saturation measurements of backpressure.

1.5 Related Work

Distributed identifier schemes have evolved considerably since the original UUID specification. RFC 9562 (Davis, Peabody & Leach, 2024) standardized UUID versions 1, 3–8, with Version 7 introducing timestamp-ordered UUIDs that address the B-tree fragmentation problems of random Version 4 UUIDs. UUID Version 7 uses a 48-bit Unix timestamp prefix followed by random bits, providing millisecond-precision ordering and global uniqueness.

The comparison includes the alternatives examined during the development of RFC 9562 including Snowflake, ULID, KSUID, and the original CUID (Davis, Peabody & Leach, 2024, sec. 2.1). These schemes illustrate different choices in identifier size, ordering, timestamp precision, and generator metadata. We include CUID2 as CUID's successor, UUID V4 and V7 as standardized random and time-ordered baselines, and database sequences as the compact storage baseline. This selection covers relevant design trade-offs without reproducing the RFC's full survey.

Twitter's Snowflake architecture (Twitter Engineering, 2010, 2012) pioneered timestamp-prefixed, worker-partitioned 64-bit identifiers for high-throughput systems. Snowflake identifiers embed a 41-bit timestamp with millisecond precision, a 10-bit machine identifier (5-bit datacenter ID and 5-bit worker ID), and a 12-bit sequence number, enabling approximately 4,096 identifiers per millisecond per machine. Its custom epoch starts from November 4, 2010 and extends to July 10, 2080 (~69.68 years).

ULIDs (Feerasta, 2019) provide a 128-bit value composed of a 48-bit timestamp with millisecond precision. Its Unix epoch starts from January 1, 1970 and extends to ~10889 CE (~8,919 years). ULIDs are encoded as 26-character Crockford Base32 strings that sort lexicographically by creation time, targeting environments where string-based identifiers are standard.

The original CUID (Elliott, 2023) was a collision-resistant identifier specification that used a k-sortable, timestamp-prefixed structure. It was deprecated by its author due to security concerns. The CUID deprecation notice warns that "all monotonically increasing (auto-increment, k-sortable), and timestamp-based ids share the security issues with Cuid" and further states that "UUID V6-V8 are also

insecure because they leak information which could be used to exploit systems or violate user privacy” (Elliott, 2023). We note that this warning is valuable but too broad for UUIDv8. Whether UUIDv8 shares these security issues depends on its application-defined payload (Davis, Peabody & Leach, 2024, sec. 5.8).

CUID2, the successor to CUID, takes a fundamentally different approach. The CUID2 authors observe that “insecure ids can cause problems in unexpected ways, including unauthorized user account access, unauthorized access to user data, and accidental leaks of user’s personal data which can lead to catastrophic effects, even in innocent-sounding applications like fitness run trackers” (Elliott, 2026). CUID2 intentionally removed timestamps from the identifier for security reasons and instead recommends a separate indexed `createdAt` column for time-based sorting. CUID2 generates identifiers by using independent entropy sources then hashing the concatenation with SHA3. This produces identifiers with strong collision resistance but relies on probabilistic uniqueness rather than deterministic construction.

KSUIDs (Segment, 2023) (K-Sortable Unique Identifier) provide a 160-bit (20-byte) binary value composed of a 32-bit timestamp. It has 1-second precision, a custom epoch from May 13, 2014 to June 19, 2150, and a 128-bit cryptographically random payload. KSUIDs also have a 27-character Base62 representation. Both binary and text representations sort lexicographically by their embedded second-resolution timestamps. Identifiers generated within the same second are ordered by their random payloads. The 128-bit random payload provides stronger collision resistance than UUID V4’s 122 random bits.

None of these approaches offers integrity verification. Table 1 presents a detailed feature comparison.

Table 1: Feature comparison of identifier schemes.

Feature	Source-Known IDs	UUID V4	UUID V7	Snowflake	ULID	CUID2	KSUID	DB Sequence
Chronological ordering	250ms	No	ms	ms	ms	No	second	Numeric
Embedded timestamp	Yes	No	Yes	Yes	Yes	No	Yes	No
Origin metadata embedding	Yes	No	No	Partial	No	No	No	No
Multi-century addressing	Yes (SKEID)	N/A	Yes	No	Yes	N/A	Partial	N/A
Entity type	Yes (8-bit)	No	No	No	No	No	No	No
Integrity check	Yes (BLAKE3 MAC)	No	No	No	No	No	No	No
Confidentiality	Encryption (AES-256)	Randomness	No	No	No	Hashing	No	No
Zero-lookup verification	Yes	No	No	No	No	No	No	No
DB primary key size	8 B (SKID)	16 B	16 B	8 B	16 B	24 chars	20 B	4/8 B
External ID size	16 B (Secure SKEID)	16 B	16 B	8 B	26 chars	24 chars	27 chars	4/8 B
UUID compatible	Yes (SKEID, UUID V8)	Yes	Yes	No	No	No	No	No

2 METHODS

Portions of this text were previously published as part of an arXiv preprint (Kılıç, 2026a).

The proposed identifier system is evaluated through specification analysis, implementation tests, performance measurement, and threat analysis. The evaluation examines representation size and ordering, identity preservation and persistence behavior, execution time and managed allocation, sequence-capacity behavior, and security properties under the stated deployment and cryptographic assumptions.

Performance evaluation uses microbenchmarks of generation, parsing, and conversion, compares generation costs with UUID versions 4 and 7, and examines backpressure through separate saturation measurements. Database-engine performance is not evaluated, and the security analysis does not constitute a formal compositional proof.

2.1 Three-Tier Identity Model

Source-Known Identifiers represent entity identity through a three-tier identity model that spans database storage, trusted internal communication (in-process or inter-process), and external access.

1. **Database Tier:** A 64-bit Source-Known ID (SKID) serves as the primary key. It embeds a 250-millisecond-precision timestamp, application topology (application identifier, instance identifier), and a per-entity-type sequence counter, providing natural ordering and compact B-tree indexing. Entity-type and epoch context are supplied by the surrounding schema and configuration, for example through the table name and the dataset’s configured epoch.

263 The 32-bit timestamp field stores unsigned 250-millisecond ticks elapsed since the start of the
264 current epoch half. Sub-second ordering at 250ms granularity eliminates coarse-grained temporal
265 ambiguity while preserving throughput within a compact 64-bit layout.

266 **2.2.2 Sequence Capacity**

267 For each application/entity-type pair on one instance, the 18-bit counter provides a **sequence capacity**
268 of $2^{18} = 262,144$ identifiers per tick. At four ticks per second, this permits $4 \times 262,144 = 1,048,576$
269 identifiers per second. Across an application's 64 instances, the **aggregate sequence capacity** for that
270 pair is $64 \times 1,048,576 = 67,108,864$ identifiers per second.

271 Implementations must reset the sequence when the timestamp advances. If the sequence capacity is
272 exhausted within a tick, the generator must wait for the next tick before issuing new identifiers for that
273 application/entity-type pair on that instance, applying backpressure to maintain uniqueness guarantees.

274 **2.2.3 Generation Algorithm**

275 The SKID generation procedure operates as follows. Given entityType, appId, appInstanceId, and
276 a configured epoch:

- 277 1. Compute elapsedTicks as the floor of 250-millisecond ticks elapsed since the epoch (four ticks
278 per second).
- 279 2. Assert that the elapsed ticks are within the epoch range (0 to $2^{33} - 1$ ticks, covering both epoch
280 halves).
- 281 3. Determine the epoch half:
 - 282 1. If $\text{elapsedTicks} < 2^{32}$, the current half is the first half (sign bit = 1, producing a negative
283 SKID). Otherwise, the current half is the second half (sign bit = 0, producing a nonnegative
284 SKID).
 - 285 2. Compute $\text{timestamp} = \text{elapsedTicks} \bmod 2^{32}$ (equivalently, $\text{elapsedTicks} \text{ AND } 2^{32} - 1$).
- 287 4. Obtain the next sequence value from the per-entity-type atomic counter (resetting on timestamp
288 change).
- 289 5. Pack fields into a 64-bit signed integer: sign bit at position 63, timestamp at positions 62–31,
290 appId at positions 30–24, appInstanceId at positions 23–18, and sequence counter at positions
291 17–0.
- 292 6. Return the packed value.

293 **2.2.4 Clock Drift Protection**

294 The system handles backward time jumps at two levels. Minor drifts freeze the timestamp until the wall
295 clock catches up, allowing the sequence counter to continue advancing within the frozen tick. The
296 reference implementation uses a 5-second freeze threshold. Drifts at or beyond this threshold are
297 considered critical. Subsequent generation requests fail, and application shutdown is requested.
298 Restarting the application and assigning a new app instance ID are responsibilities of an external
299 configuration or coordination service (see Limitation 2). This mechanism maintains nondecreasing
300 generation timestamps within a running process. A new app instance ID is used to prevent collisions
301 across application restarts and does not preserve chronological monotonicity across a restart after clock
302 rollback.

2.3 Source-Known Entity ID (SKEID) 128-bit Specification

2.3.1 Byte Layout

An SKEID occupies 16 bytes (128 bits), structured as shown in Figure 3.

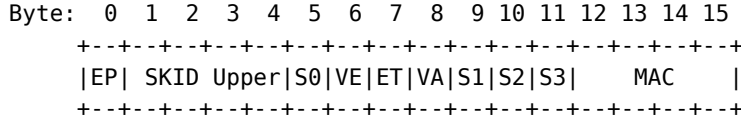


Figure 3: SKEID 16-byte field layout (big-endian, RFC 9562 network byte order).

Table 3 details the byte-level layout of the SKEID fields.

Table 3: SKEID byte layout.

Field	Byte(s)	Width	Purpose
Epoch	0	8 bits	Epoch index (highest lexicographic priority)
SKID upper half	1–4	32 bits	SKID bits 63–32, sign-toggled for lexicographic order
SKID low byte 0	5	8 bits	MSB of SKID lower half (timestamp LSB, app ID)
Version marker	6	8 bits	0x8D (UUID V8, RFC 9562 Section 5.8 octet 6)
Entity Type	7	8 bits	Domain entity classification
Variant marker	8	8 bits	0x8D (RFC 9562 Section 4.1 octet 8 variant)
SKID low bytes	9–11	24 bits	Remaining SKID lower half
MAC	12–15	32 bits	BLAKE3 keyed MAC

All fields are encoded in big-endian (network byte order) per RFC 9562. The epoch byte occupies byte 0 to ensure that higher epoch values sort lexicographically after lower epoch values, regardless of timestamp. The upper SKID half at bytes 1–4 is sign-toggled (XOR with 0x80000000) before encoding. This converts the signed SKID sort order (negative before nonnegative) to unsigned byte sort order, preserving chronological ordering in lexicographic comparisons across both epoch halves. The most significant byte of the lower SKID half occupies byte 5, immediately after the upper half, ensuring that the timestamp least-significant bit and leading application topology bits participate in lexicographic comparison before the entity type at byte 7. The remaining lower SKID half is split around the variant marker at byte 8, with bytes 9–11 holding the last three bytes. Since the version marker, entity type, and variant marker are constant for all plain SKEIDs of a given type, lexicographic comparison of the split lower half operates correctly.

The marker bytes at positions 6 and 8 serve dual purposes.

1. **UUID V8 compatibility** makes the SKEID a valid RFC 9562 UUID.
2. **Detection signal** enables the parser to distinguish plaintext SKEIDs from encrypted Secure SKEIDs.

2.3.2 MAC Computation

Integrity verification relies on BLAKE3 keyed MAC (O'Connor et al., 2021). The computation procedure clears the four MAC bytes (positions 12–15) to zero, then computes a 4-byte BLAKE3 keyed MAC over the full 16-byte buffer. The resulting 4 bytes are written into the contiguous MAC positions (bytes 12–15).

BLAKE3 was selected for performance and security margin. In the BenchmarkDotNet measurements, BLAKE3 keyed hashing with a reused hasher averaged 81.03 ns on 16-byte inputs, compared with 306.75 ns for HMAC-SHA-256, approximately 3.79 times faster (see supplementary file `supplementary-hash-benchmark-report.html`).

Generated MAC is truncated to 32 bits, offering a 2^{-32} success probability per guessed tag. This truncation is acceptable within the defense-in-depth architecture where the MAC is one of multiple verification layers, not the sole security mechanism (see MAC Truncation Analysis).

The epoch byte (byte 0) is not cleared before MAC computation. It participates in the MAC input, adding tamper-resistance for the epoch field.

2.3.3 Generation Algorithm

Given a 64-bit SKID, epoch, entity type, and MAC key, the SKEID generation procedure:

1. Allocate a 16-byte buffer initialized to zero.

2. Split the SKID into upper and lower 32-bit halves. Toggle the most significant bit of the upper half (XOR with 0x80000000) and write it in big-endian to bytes 1–4.
3. Write the most significant byte of the SKID lower half to byte 5.
4. Write the epoch byte to byte 0, the version marker 0x8D to byte 6 (RFC 9562 Section 5.8 octet 6), entity type to byte 7, and variant marker to byte 8 (RFC 9562 Section 4.1 octet 8).
5. Write the remaining three bytes of the SKID lower half to bytes 9–11.
6. Compute the BLAKE3 keyed MAC and write it to bytes 12–15.
7. Construct the UUID from the 16-byte buffer, interpreting all fields in big-endian (network byte order) per RFC 9562, and return the result.

2.3.4 Parsing Algorithm

Given a 16-byte UUID `entityId` and MAC key K_{mac} :

1. Convert the UUID to a 16-byte big-endian byte array (network byte order, per RFC 9562).
2. Check that the epoch byte at position 0 represents a supported epoch (`guidBytes[0] <= MaxSupportedEpoch`) and check for exact primary marker bytes (0x8D at position 6, 0x8D at position 8) in the byte array. If the epoch or markers do not match, return INVALID.
3. Extract the 32-bit MAC from bytes 12–15 and clear bytes 12–15 to zero.
4. Compute the 4-byte BLAKE3 keyed MAC over the 16-byte buffer using K_{mac} . If it does not match the extracted MAC, return INVALID.
5. Extract the epoch byte (byte 0) and entity type (byte 7). Reconstruct the 64-bit SKID from the upper half (bytes 1–4, reversing the sign-bit toggle with 0x80000000) and lower half (byte 5 and bytes 9–11).
6. Return the extracted SKID, epoch, and entity type as a valid plaintext SKEID.

2.4 Secure SKEID 128-bit Specification

2.4.1 AES-256-ECB Single-Block Encryption

Table 4 shows the Secure SKEID byte layout after AES-256 encryption.

Table 4: Secure SKEID byte layout.

Field	Byte(s)	Width	Purpose
Ciphertext	0–15	128 bits	AES-256 encrypted SKEID (pseudorandom bytes)

As a single AES block, the entire SKEID is encrypted. No internal structure is visible to external consumers. Identical SKEID plaintexts encrypted under the same key produce identical ciphertexts, allowing repeated identifiers to be recognized.

This conversion has zero expansion. A 16-byte SKEID becomes a 16-byte Secure SKEID. By comparison, AES-SIV supports deterministic authenticated encryption but adds 16 bytes of overhead, producing 32 bytes for a 16-byte plaintext (Harkins, 2008, sec. 2.6 and 4). The AES-GCM profiles in RFC 5116 likewise add a 16-byte authentication tag, excluding any separately transmitted nonce (McGrew, 2008, sec. 5.1 and 5.2). Although NIST SP 800-38D permits several tag lengths, all require additional space, so the complete authenticated output still expands (Dworkin, 2007, sec. 5.2.1.2). Neither construction can therefore reversibly carry a SKEID in a self-contained 128-bit output. Zero expansion does not imply security equivalent to AES-SIV or AES-GCM, or preservation of UUID version and variant bits (see Limitations 4 and 6).

A Secure SKEID is produced by encrypting the entire 16-byte SKEID plaintext using AES-256 (National Institute of Standards and Technology, 2023) as a single-block cipher. AES has undergone extensive public cryptanalysis, reviewed in NIST IR 8319 (Mouha, 2021). Under the pseudorandom permutation (PRP) assumption with a uniformly chosen key, AES-256 applied to a single 128-bit block is computationally indistinguishable from a uniformly random permutation. A pseudorandom function (PRF) is similarly computationally indistinguishable from a uniformly random function. The birthday threshold for distinguishing a random permutation from a random function is approximately $2^{n/2}$ queries (Bellare, Krovetz & Rogaway, 1998, sec. 2.2). For AES with $n = 128$, candidate ciphertexts for distinct plaintexts can be modeled approximately as independent, uniformly random blocks as long as the query count under a single key remains well below 2^{64} .

393 Because SKEID encryption always operates on exactly one block, the multi-block weakness of ECB
 394 mode does not apply. For a single block, ECB is mathematically identical to Cipher Block Chaining
 395 (CBC) with a zero initialization vector, as shown in Equation (1):

$$C = \text{AES}_{K_{\text{aes}}}(P \oplus \mathbf{0}) = \text{AES}_{K_{\text{aes}}}(P), \quad (1)$$

396 where P is the plaintext block and $\mathbf{0}$ is the all-zero 128-bit block. No nonce is required, and there is
 397 no nonce-reuse vulnerability.

398 Quantum attacks on AES via Grover's algorithm reduce the effective security of AES-256 to
 399 approximately 128 bits, which still remains computationally infeasible (Bonnetain, Naya-Plasencia &
 400 Schrottenloher, 2019).

401 **2.4.2 Key Separation**

402 The three-tier identity system uses separate MAC and encryption keys for each key-ring entry. A MAC
 403 key is used for BLAKE3 keyed MAC (integrity verification) and an AES key is used for AES-256-ECB
 404 (confidentiality). The two keys must not be the same value. Implementations should derive both keys
 405 from a single master secret using a key derivation function with domain separation, such as BLAKE3's
 406 `derive_key` mode with distinct, hardcoded, globally unique context strings identifying the application
 407 and key purpose (O'Connor et al., 2021, sec. 6.2).

408 **2.4.3 Generation Algorithm**

409 Given a 64-bit SKID, epoch, entity type, and key pair $(K_{\text{mac}}, K_{\text{aes}})$:

- 410 1. Construct the 16-byte plaintext SKEID buffer using the SKEID generation algorithm with default
 411 variant marker `0x8D`, and compute its MAC using K_{mac} .
- 412 2. Encrypt the 16-byte buffer using AES-256-ECB with K_{aes} to produce candidate ciphertext.
- 413 3. Check whether the ciphertext triggers the collision guard (Collision Guard Mechanism below): if
 414 the ciphertext exhibits a supported epoch, exact markers `0x8D` at byte 6 and `0x8D` at byte 8, and a
 415 valid MAC under any enabled verification key, regenerate with successive variant bytes from
 416 `0x8E` through `0xBF` (recomputing the MAC and re-encrypting with the same key pair) until a
 417 non-colliding ciphertext is obtained.
- 418 4. Construct the UUID from the final 16-byte ciphertext in big-endian order and return the Secure
 419 SKEID.

420 **2.4.4 Collision Guard Mechanism**

421 Without the collision guard, there exists a combined approximately 2^{-48} probability that ciphertext
 422 coincidentally matches both the plaintext marker bytes and produces a valid MAC when interpreted as a
 423 plaintext SKEID. This would cause a Secure SKEID to be misclassified as a plaintext SKEID with
 424 incorrect data.

425 The collision guard eliminates this edge case for the verification keys enabled during generation.
 426 When the ciphertext passes plaintext marker, epoch, and MAC checks under any enabled key, the
 427 plaintext is regenerated with successive variant bytes from `0x8E` through `0xBF` (50 alternatives). Each
 428 candidate is authenticated and encrypted with the same generation key pair. Under the PRP assumption,
 429 each variant produces distinct pseudorandom ciphertext. Termination is deterministic. The loop is
 430 bounded at 51 total attempts (1 primary + 50 alternatives), with an implementation-defined error on
 431 exhaustion (the reference implementation throws `JackpotException`). For a single verification key, the
 432 idealized estimate for exhausting all 51 attempts is approximately $2^{-48 \times 51}$. Multiple enabled keys
 433 increase the per-candidate collision probability.

434 **2.4.5 Parsing Algorithm**

435 Given a 16-byte UUID `entityId` and key pair $(K_{\text{mac}}, K_{\text{aes}})$:

- 436 1. Convert the UUID to a 16-byte big-endian byte array (network byte order, per RFC 9562).
- 437 2. Decrypt the 16-byte block using AES-256-ECB with K_{aes} .
- 438 3. Check the decrypted plaintext for markers and supported epoch: `guidBytes[0] <=`
 439 `MaxSupportedEpoch`, `guidBytes[6] == 0x8D` (exact version match), and `(guidBytes[8] &`
 440 `0xC0) == 0x80` (RFC 9562 §4.1 variant range `0x80–0xBF`). If any check fails, return `INVALID`.

- 441 4. Verify the decrypted plaintext's MAC using K_{mac} (extracting bytes 12–15, zeroing the slots, and
442 recomputing the MAC). If verification fails, return INVALID.
- 443 5. Recover the variant byte $V = \text{guidBytes}[8]$. If $V < 0x8D$, return INVALID without further key
444 fallback.
- 445 6. If $V > 0x8D$, perform backward collision-guard verification as specified in the Backward
446 Verification Algorithm section. If verification fails, return INVALID without further key fallback.
- 447 7. Extract the epoch, entity type, and SKID, and return the result as a valid Secure SKEID (Secure
448 `== true`).

449 **2.4.6 Backward Verification Algorithm**

450 During parsing, the recovered candidate must first pass its own MAC verification. When a non-default
451 variant byte V is recovered from the decrypted plaintext ($0x8D < V \leq 0xBF$), the parse algorithm must
452 verify every preceding variant from $V - 1$ down to $0x8D$, inclusive. For each candidate, it reconstructs
453 the SKEID with the same SKID, epoch, and entity type, recomputes its MAC, encrypts it with the
454 authenticated key pair, and checks the resulting ciphertext for a plaintext collision under any enabled
455 verification key. The reconstruction key pair remains fixed throughout the proof. The parser rejects the
456 identifier as soon as any preceding candidate does not collide, without retrying the identifier under
457 another key.

458 This verifies the complete preceding collision history directly, without assuming that untrusted input
459 was produced by the generator. For example, accepting variant $0x8F$ requires collisions at both $0x8E$
460 and $0x8D$. A collision at $0x8E$ alone is insufficient. Verification requires at most 50 preceding-candidate
461 checks.

462 For the default variant ($0x8D$), parsing performs no predecessor checks. Under the idealized
463 single-key estimate, generation requires variant escalation with probability approximately 2^{-48} per
464 identifier. Epoch filtering can reduce this probability, while additional enabled verification keys can
465 increase it.

466 A successfully generated Secure SKEID cannot be accepted as a plaintext SKEID, provided that
467 generation and parsing use the same verification keys, epoch-acceptance policy, and
468 plaintext-acceptance checks.

469 **2.4.7 Auto-Detection Parse Logic**

470 Applications operating across both trusted internal and external boundaries can use auto-detection
471 parsing to accept both plaintext and encrypted identifiers:

- 472 1. Convert the UUID to a 16-byte big-endian byte array (network byte order, per RFC 9562).
- 473 2. If the byte array exhibits a supported epoch (`guidBytes[0] <= MaxSupportedEpoch`) and exact
474 primary markers ($0x8D$ at position 6, $0x8D$ at position 8), attempt plaintext verification using the
475 Plain SKEID Parsing Algorithm. If MAC verification succeeds, return as a plaintext SKEID.
- 476 3. Otherwise, attempt secure parsing using the Secure SKEID Parsing Algorithm by decrypting with
477 K_{aes} and validating markers, MAC, and backward collision proofs.
- 478 4. If neither path succeeds, return INVALID.

479 The probability of a random or encrypted byte sequence coincidentally containing the two marker
480 bytes $0x8D8D$ at the exact positions is $2^{-16} = 1/65,536$. Requiring one specific 8-bit epoch further
481 reduces this coincidental match probability to $2^{-(16+8)} = 2^{-24} = 1/16,777,216$. If plaintext MAC
482 verification fails, parsing falls back to the decryption path. For generated Secure SKEIDs, the collision
483 guard ensures that plaintext acceptance fails.

484 Consequently, endpoints exposed to untrusted networks must avoid auto-detection or explicitly
485 require the encrypted representation. Because plaintext SKEIDs expose internal metadata (timestamps,
486 entity types, topology, and sequence counters) in fixed byte positions, accepting them at external
487 boundaries permits structure-based enumeration. An attacker could then craft structured candidates and
488 test guessed MACs directly against the 32-bit truncation bound, bypassing the confidentiality and
489 anti-enumeration guarantees of AES-256 encryption. In the reference implementation, callers enforce
490 this boundary by selecting `SourceKnownEntityIdFormat.Secure` rather than `Auto`.

491 **2.4.8 Tier Conversion**

492 Implementations support bidirectional conversions between tiers (Figure 1) for already-validated
493 identifiers:

- **ToSecure:** Converts a plaintext SKEID to a Secure SKEID by invoking the Secure SKEID generation algorithm. If the identifier is already secure, it is returned unchanged.
- **ToPlain:** Converts a Secure SKEID to a plaintext SKEID. When converting an encrypted identifier whose collision guard escalated the variant byte (0x8E–0xBF), ToPlain does not expose the raw decrypted block (which would fail standard plaintext marker parsing). Instead, it canonicalizes the identifier by reconstructing the canonical plaintext SKEID with the default variant byte 0x8D and a freshly computed BLAKE3 keyed MAC under K_{mac} . If the identifier is already plaintext, it is returned unchanged.

2.4.9 Numeric Walkthrough

This example reproduces the public test vector from Appendix A of the draft specification (Kılıç, 2026b). The collision-guard scenario that follows is hypothetical and is not triggered by this vector.

Setup: Consider a SKID with value 0x881B3200050C002A (signed decimal -8639256484514103254), timestamp = 272,000,000 ticks from epoch (68,000,000 seconds at 250ms precision), app ID = 5, app instance ID = 3, sequence = 42, entity type 0x01, and epoch 0x00 beginning on 2025-01-01T00:00:00Z.

The public test key material and derived keys are shown below. Each hexadecimal string represents 32 raw bytes, not UTF-8 text. Implementations can use the displayed derived keys directly to reproduce the wire vector in tests. These public test keys must not be used in production.

The public master-key value also appears as the 256-bit key-encryption key in the test vector in RFC 3394 (Schaad & Housley, 2002, sec. 4.3). The MAC and AES keys shown below are derived from this master key by the reference implementation.

Master key:

000102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F

MAC key (K_{mac}):

5E293E89136745A96B70EB8C8F81CDFCAED177BE5358BC83D3039FB6607FD8FE

AES key (K_{aes}):

4988F97FF724CD086BDFEC83497C3527B3656F35F0911BEEAA6BCE4BB92D3BC7

Step 1 SKEID Construction: The 16-byte SKEID buffer is constructed in big-endian order. Byte 0 receives the epoch (0x00), bytes 1–4 receive the sign-toggled SKID upper half (0x881B3200 XOR 0x80000000 = 0x081B3200), byte 5 receives 0x05, byte 6 receives the version marker (0x8D), byte 7 receives the entity type (0x01), byte 8 receives the default variant marker (0x8D), and bytes 9–11 receive 0x0C002A. With bytes 12–15 initially zero, the first four BLAKE3 keyed output bytes are 492C0E75. Writing this MAC into bytes 12–15 produces:

SKEID bytes: 00 08 1B 32 00 05 8D 01 8D 0C 00 2A 49 2C 0E 75

SKEID UUID: 00081b32-0005-8d01-8d0c-002a492c0e75

Step 2 Encryption and Collision Check: Encrypting the 16-byte plaintext with AES-256-ECB using K_{aes} and no padding produces ciphertext C_1 :

Secure bytes: 65 20 68 A4 36 12 CC 4B 8A BB 83 B8 53 DC 67 86

Secure UUID: 652068a4-3612-cc4b-8abb-83b853dc6786

Ciphertext bytes 6 and 8 are 0xCC and 0x8A, so the plaintext marker checks fail and the collision guard accepts C_1 without variant escalation. Parsing either representation recovers SKID -8639256484514103254 and entity type 1. With the same keys, ToSecure and ToPlain reproduce the two UUIDs above.

Hypothetical Collision Scenario: For a different input or key pair, suppose the first ciphertext candidate C_1 has a supported epoch byte, contains 0x8D at positions 6 and 8, and has a valid BLAKE3 keyed MAC at positions 12–15 when interpreted as a plaintext SKEID. The MAC is computed over a copy of the full 16-byte buffer of C_1 with bytes 12–15 set to zero. Conditional on the epoch byte being supported, this marker-plus-MAC coincidence has probability $\approx 2^{-48}$ for one verification key.

When the collision guard activates, the plaintext variant byte (position 8) is changed from 0x8D to 0x8E. The BLAKE3 MAC is recomputed over the modified plaintext. The new plaintext is encrypted with the same K_{aes} , producing ciphertext C_2 . Under the ideal-permutation approximation, the next distinct candidate's collision probability, including the single accepted epoch, is approximately 2^{-56} . If C_2 passes the check, it becomes the final Secure SKEID; otherwise, generation continues with the next variant.

Backward Verification: When a consumer parses C_2 by decrypting with K_{aes} , the recovered plaintext reveals variant byte 0x8E ($> 0x8D$). The parser must verify that the escalation was legitimate.

1. Reconstruct the SKEID with variant 0x8D (replacing 0x8E and recomputing the MAC).
2. Encrypt that reconstruction with K_{aes} to obtain C_1 .
3. Check that C_1 has a supported epoch, matching markers, and a valid MAC, confirming the collision that justified the escalation.
4. For recovered variant 0x8E, 0x8D is the only preceding candidate, so this check covers the complete chain. Higher recovered variants require checking every preceding candidate down to 0x8D.

If the backward check fails (the reconstructed C_1 does not exhibit the coincidence), the parser rejects the identifier as invalid. This prevents an attacker from crafting identifiers with arbitrary variant bytes.

`IetfDraftTestVectorTests` asserts the master-key derivation, SKID, MAC, plaintext and ciphertext bytes, UUID strings, and round trips shown above.

2.5 Epoch and Extensibility

SKEID byte 0 carries an 8-bit epoch index. Each value selects a 2^{31} -second window (approximately 68 years), with epoch zero beginning on 2025-01-01. The 256 epoch values provide approximately 17,421 years of total coverage.

Table 5 illustrates epoch addressing across the full epoch range.

Table 5: Epoch addressing.

Epoch Value	Start	End
0x00	January 1, 2025	January 19, 2093
0x01	January 19, 2093	February 7, 2161
⋮	⋮	⋮
0xFF	19378	19446

This two-level time addressing gives the identity system a multi-century lifespan without identifier collisions or sorting degradation. Epoch boundaries, total coverage, half-epoch boundary year, and sign-bit sort-order invariants in Table 5 are verified by the `PaperEpochAddressabilityTests` unit test suite in the reference implementation.

2.6 Defense-in-Depth Security Architecture

The three-tier identity system employs multiple verification and defence layers. An attacker attempting to forge a valid identifier must satisfy all applicable validation checks. AES-256 provides confidentiality and prevents structure-based enumeration under the PRP assumption. However, it is not an additional independent forgery-probability factor.

Table 6 enumerates the defense-in-depth security layers and their contributions to rejecting uniformly random decrypted blocks.

Table 6: Defense-in-depth security layers.

Layer	Mechanism	Random-block acceptance contribution
1. AES-256 Encryption	Confidentiality with PRP over 128 bits	Prevents structure-based enumeration
2. BLAKE3 Keyed MAC	32-bit truncated	2^{-32}
3. Marker Detection	0x8D8D at fixed positions (default variant)	2^{-16}
4. Epoch Match	Must match one expected 8-bit epoch or range	2^{-8} for single accepted epoch
5. Entity Type Match	Must match one expected 8-bit entity type	2^{-8} optional
6. App ID Match	Must match one expected 7-bit app ID	2^{-7} optional

Layer	Mechanism	Random-block acceptance contribution
7. Existence Probability	Forged SKID must reference a record	Deployment dependent
8. Rate Limiting	Restricts attempts per time period	Implementation defined

For a uniformly random decrypted block, the default-marker, epoch, entity-type, app ID, and MAC checks have independent acceptance events. This allows their acceptance probabilities to be multiplied. For each key pair, all five validation checks leave 2^{57} plaintext blocks out of 2^{128} . AES permutes these blocks without changing their number.

For a secure SKEID-only endpoint validating against one key pair in the key-ring, if validation checks only the 32-bit keyed MAC and the two fixed marker bytes, the default-variant acceptance probability is $2^{-(32+16)} = 2^{-48}$ per uniformly random ciphertext attempt. Requiring one specific 8-bit epoch reduces it to 2^{-56} . Requiring one specific 8-bit entity type further reduces it to 2^{-64} , and one specific 7-bit app ID reduces it to 2^{-71} .

The probabilities above apply specifically to the default-variant path. Every non-default variant also requires backward collision-guard verification. Every reconstructed predecessor ciphertext, down to variant 0x8D, must satisfy the plaintext marker, epoch, and MAC checks. This further restricts acceptance beyond the current variant's own validation checks.

Record existence checks can reduce acceptance further. Accepting multiple key pairs, entity types, app IDs, or epochs requires adjusting the default-variant acceptance probability. These estimates concern random-ciphertext guessing, not replay or a formal compositional security proof (see Limitation 4). At endpoints accepting plaintext SKEIDs, an attacker can set the public fields directly, so these field checks do not extend the 32-bit MAC forgery bound.

2.6.1 MAC Truncation Analysis

The following comparison applies the HMAC guidance to BLAKE3 Keyed MAC and default-variant validation checks solely by analogy. It does not make the 32-bit BLAKE3 MAC equivalent to a 64-bit MAC or establish compliance with NIST HMAC guidance.

The SKEID 32-bit MAC truncation matches the minimum MacTag length specified for HMAC using approved hash algorithms in NIST SP 800-107 (Dang, 2012, sec. 5.3.3). Section 5.3.5 of the same document also discourages MacTags shorter than 64 bits. However, the 128-bit SKEID format constrains the space available for the MAC field, as the remaining bits carry the SKID, entity type, epoch, and marker bytes. Under the conditions stated in Defense-in-Depth Security Architecture, the probability that a uniformly random ciphertext passes all Secure SKEID default-variant validation checks is 2^{-71} , lower than the per-attempt random-tag guessing probability of 2^{-64} for a 64-bit MacTag.

Section 5.3.5 of the same document provides the general framework for assessing truncated MAC forgery risk. For a λ -bit MacTag and an attempt budget of $q = 2^t$, Equation (2) gives the idealized tag-guessing bound:

$$\Pr[\text{forge}] \leq \min(1, q \cdot 2^{-\lambda}) = \min(1, 2^{t-\lambda}). \quad (2)$$

Applying this estimate by analogy to the 32-bit MAC in plaintext SKEIDs ($\lambda = 32$): if a system permits 2^{12} (4,096) failed verification attempts before retiring that MAC key from verification, the forgery likelihood is bounded by 2^{-20} , approximately one in a million, under the idealized tag-guessing model. At 100 rate-limited attempts per second, this 2^{12} budget is exhausted in roughly 41 seconds.

Secure SKEIDs use AES-256 encryption under the PRP assumption to prevent structure-based enumeration. At endpoints that reject plaintext SKEIDs, encryption also prevents direct testing of guessed plaintext MAC values. Random ciphertext guesses remain subject to validation and rate limiting.

In December 2022, NIST announced plans to withdraw SP 800-107 Rev. 1 (NIST, 2022). The 2024 Initial Public Draft of NIST SP 800-224 (Turan & Brandão, 2024) includes guidance on truncated HMACs. Its recommendations inform this BLAKE3 analysis by analogy.

2.7 Key Rotation and Key-Ring Fallback

A key-ring containing one or more key entries is required. Exactly one key is designated as the default (active) key at any time. During routine rotation, applications may retain previous keys for parse fallback. All new SKEID and Secure SKEID generation uses the current default key. The reference

616 implementation accepts only the default key unless the application explicitly enables
617 `UseSourceKnownIdKeyFallback` (default `false`). With fallback enabled, parsing attempts verification
618 with the current default key first, then the remaining keys in configured order. Once a candidate
619 authenticates, its key pair remains fixed for backward verification. A failed proof rejects the identifier.

620 Key rotation does not require re-encrypting existing database records because the 64-bit SKID stored
621 in the database contains no cryptographic material. Given the SKID and entity type, a new SKEID or
622 Secure SKEID can be regenerated with the new key at any time. In the event of key compromise:

- 623 1. Set a fresh, uncompromised key as the default.
- 624 2. Remove the compromised key from verification on every instance. Disabling fallback rejects all
625 non-default keys, while removing the compromised entry allows other trusted fallback keys to
626 remain enabled.
- 627 3. Reissue affected external identifiers from trusted 64-bit SKIDs and entity types as needed.
628 Previously issued identifiers that require the removed key no longer authenticate, while the stored
629 64-bit SKIDs remain unchanged.

630 The application owns key retention and rollout policy. Retaining a compromised key for
631 compatibility also retains the attacker's ability to forge accepted identifiers.

632 2.8 Reference Implementation

633 The reference implementation is provided as part of the open-source `DRN.Framework` (Kılıç, 2026c),
634 organized across four NuGet packages.

635 The current implementation generates and accepts only encoded epoch 0. Support for nonzero
636 epochs and operational epoch transitions remains future work.

- 637 • **`DRN.Framework.SharedKernel`** defines interfaces (`ISourceKnownEntityIdOperations`),
638 base classes (`SourceKnownEntity`), and parsed identity structs (`SourceKnownId`,
639 `SourceKnownEntityId`).
- 640 • **`DRN.Framework.Utills`** implements cryptographic operations, timestamp management, sequence
641 management (`SequenceManager<TEntity>` with lock-free atomic operations), and bit packing
642 (`SourceKnownIdUtills`, `SourceKnownEntityIdUtills`).
- 643 • **`DRN.Framework.EntityFramework`** implements persistence integration including
644 `SourceKnownIdValueGenerator` (EF Core value generator for automatic SKID assignment),
645 `DrnSaveChangesInterceptor` (auto-assigns SKID and SKEID on insert),
646 `DrnMaterializationInterceptor` (reconstructs full SKEID from stored SKID on database
647 read), and `SourceKnownRepository<TContext, TEntity>` (generic repository with SKEID
648 validation and cursor-based pagination).
- 649 • **`DRN.Framework.Testing`** implements integration and unit test infrastructure that provides
650 Testcontainers orchestration for ephemeral PostgreSQL instances, auto-mocking through data
651 attributes (`[DataInline]`, `[DataInlineUnit]`), and convention-based test contexts
652 (`DrnTestContext`, `DrnTestContextUnit`).

653 The specified identifier structure and tier transformations provide a common contract across domain
654 models, identifier utilities, and persistence integration. The reference implementation demonstrates this
655 organization through shared contracts in `DRN.Framework.SharedKernel`, generation and parsing in
656 `DRN.Framework.Utills`, and persistence integration in `DRN.Framework.EntityFramework`.

657 The common contract also enables early validation of expected entity types and app IDs embedded
658 in the Source Known Identifiers, allowing mismatched identifiers to be rejected before database access.
659 These checks support fail-fast behavior and focused tests for incorrect entity-type or app ID usage.

660 The implementation uses attribute-based dependency injection (`[Singleton<Type>]`,
661 `[Scoped<Type>]`, `[Transient<Type>]`) and entity type attributes (`[EntityType<TApp>(byte)]` for
662 entity type and application partition registration) and integrates with Domain-Driven Design (Evans,
663 2003) patterns through the `SourceKnownEntity` abstract base class. The testing strategy and validation
664 evidence for these packages are described in the Validation and Maturity section.

665 **UUID byte-order interoperability:** The SKEID wire layout uses RFC 9562 network byte order. To
666 avoid the mixed-endian byte-array representation used by default in .NET Guid APIs, the reference
667 implementation constructs values with `Guid(ReadOnlySpan<byte>, bigEndian: true)` and
668 extracts bytes with `Guid.TryWriteBytes(guidBytes, bigEndian: true, out _)`. These
669 conversions preserve the specified byte layout across runtimes. Secure SKEIDs retain UUID string
670 formatting, but encryption does not preserve RFC version and variant bits (see Limitation 6).

671 2.9 Validation and Maturity

672 Active development of the three-tier identity system began in November 2023 as part of the
673 DRN.Framework. The framework has undergone continuous refinement across versioned releases
674 (v0.1.0 through v0.10.0). Each release is validated through integration and unit tests built on
675 DRN.Framework.Testing. Performance is measured separately with BenchmarkDotNet.

676 The framework's correctness is validated through multiple test and analysis tiers.

- 677 • **Unit tests:** In-memory tests covering SKID generation, SKEID construction with MAC
678 verification, and Secure SKEID encryption/decryption round-trips.
- 679 • **Paper-verification tests:** Dedicated suites (`PaperNumericWalkthroughTests`,
680 `PaperEpochAddressabilityTests`, `IetfDraftTestVectorTests`) assert the exact numeric
681 values, byte layouts, epoch boundaries, lexicographic ordering across epoch halves, and sort-order
682 invariants presented in this paper. Implementation changes that violate these assertions produce
683 test failures.
- 684 • **Integration tests:** Testcontainers-based tests provision ephemeral PostgreSQL instances,
685 exercising Entity Framework Core value generation, interceptors, and cursor-based pagination
686 with SKID ordering.
- 687 • **End-to-end validation:** Test applications validated behavior across the three tiers: SKID
688 assignment during entity creation, SKEID exposure within trusted application-to-application
689 communication, and Secure SKEID generation for external API responses. These applications
690 confirm that bidirectional transformations preserve data integrity across the complete round-trip.
- 691 • **Static analysis:** GitHub CodeQL and SonarCloud continuously scan the codebase for security
692 vulnerabilities, code quality issues, and potential bugs. The framework maintains zero critical or
693 high-severity findings.

694 The NuGet packages have been published on NuGet.org since the initial framework release and are
695 available for community review and adoption. They are published through an automated GitHub
696 Actions CI/CD pipeline that runs the complete test suite before each release.

697 2.10 Reference Benchmark Environment

698 The identifier-generation, parsing, and conversion benchmarks under normal load were conducted using
699 BenchmarkDotNet (Akinshin, 2025) v0.15.8 with the following hardware and software configuration.

- 700 • **Hardware:** Apple M2, 1 CPU, 8 logical and 8 physical cores
- 701 • **Operating System:** macOS 27.0 (26A428) [Darwin 27.0.0]
- 702 • **Runtime:** .NET 10.0.12 (10.0.1226.42308), Arm64 RyuJIT armv8.0-a
- 703 • **SDK:** .NET SDK 10.0.401
- 704 • **Benchmark configuration under normal load:** `OutlierMode=RemoveUpper`,
705 `InvocationCount=262,144`, `IterationCount=120`, `WarmupCount=120`, `UnrollFactor=1`
- 706 • **Benchmark configuration under saturation:** `OutlierMode=RemoveUpper`,
707 `InvocationCount=786,432`, `IterationCount=40`, `WarmupCount=40`, `UnrollFactor=1`

708 `OutlierMode` is explicitly set to `RemoveUpper`, the standard BenchmarkDotNet throughput default
709 that removes upper statistical outliers before computing summary statistics.

710 Mean, Error, and Standard Deviation are reported in nanoseconds (ns) and rounded to two decimal
711 places throughout the manuscript. Error denotes the 99.9% confidence interval (CI) margin calculated
712 by BenchmarkDotNet over iterations remaining after upper outlier removal. Speedup factors are
713 calculated by dividing the baseline mean by the operation mean using the source report values before
714 rounding the result to two decimal places.

The benchmark configuration under normal load uses 120 measurement iterations with 262,144 invocations per iteration. This invocation count matches the per-tick sequence capacity and is chosen to measure per-operation computational cost within sequence capacity under normal load. A 260-millisecond `IterationSetup` delay between iterations allows the sequence time scope to reset, so each iteration starts from a clean state.

The saturation benchmark uses `InvocationCount=786,432` (3 times the sequence cap), `IterationCount=40`, and `WarmupCount=40` (matching the iteration count), with `OutlierMode=RemoveUpper` and `UnrollFactor=1`. It is configured to observe backpressure behavior rather than to produce absolute timings directly comparable with the run under normal load. These measurements use the same hardware, operating system, runtime, and SDK listed above.

Both configurations schedule 31,457,280 measurement invocations per operation ($\text{InvocationCount} \times \text{IterationCount}$), distributing the same invocation count across different batch sizes to examine sequence-capacity backpressure.

2.11 AI-Assisted Research and Development

The following AI tools and large language model (LLM) systems were used in the research process and development of the reference implementation.

Software development, documentation, and testing: AI tools from Anthropic (Claude Opus/Sonnet series), Google (Gemini Pro/Flash series), OpenAI (o1, 4o, GPT-5.6 Sol, GPT-6 Astra), Qwen 3, and DeepSeek R1 were used across the DRN-Project development lifecycle, including the Source-Known Identifiers implementation, as agentic coding assistants for code generation, refactoring, code review, test scaffolding, and software documentation drafting. CodeRabbit was also used for AI-assisted code review. Source-Known Identifiers were added to the DRN-Project roadmap on 19 November 2023. These AI tools were progressively adopted as they became available over the subsequent two-plus years of development. All generated code, tests, and documentation were reviewed, validated, and modified by the author.

Literature research: Google Gemini Deep Research was used to assist with surveying existing distributed identifier schemes and locating relevant prior work. All identified sources were independently verified by the author before citation.

The complete research and reference implementation development history in the Git version control system is publicly available on GitHub (<https://github.com/duranserkan/DRN-Project>), providing an audit trail of commit progression and automated review interactions.

All AI-assisted outputs were reviewed, validated, and refined by the author. The architectural design, cryptographic choices, three-tier trust model, and all technical decisions are solely the work of the author. The author takes full responsibility for the correctness and originality of the work.

3 RESULTS

3.1 Performance Results

Table 7 presents the BenchmarkDotNet performance measurements for the measured operations across all three tiers.

Table 7: Generation, parsing, and conversion performance under normal load.

Operation	Mean (ns)	Error (ns)	Standard deviation (ns)
Random 64-bit integer generation	177.61	2.96	9.58
Random UUID V4 generation	361.94	1.94	6.31
Random UUID V7 generation	384.11	1.95	6.33
SKID generation	10.85	0.64	2.09
SKEID generation	207.28	3.10	10.08
Secure SKEID generation	224.62	3.90	12.67
SKID parsing	5.35	0.41	1.34
SKEID parsing	216.76	6.35	20.62
Secure SKEID parsing	250.95	3.34	10.84
ToPlain	208.46	2.28	7.39
ToSecure	221.23	2.98	9.62

Note: Error denotes the 99.9% confidence-interval margin. Statistics are calculated over measurement iterations remaining after upper outlier removal.

Table 8 presents generation performance under saturation. Each batch contains 786,432 invocations (three times the per-tick sequence capacity), with a 260-millisecond delay between batches excluded from measurement.

Table 8: Generation performance under saturation.

Operation	Mean (ns)	Error (ns)	Standard deviation (ns)
SKID generation	611.22	4.82	8.57
SKEID generation	611.23	4.99	8.88
Secure SKEID generation	611.67	6.33	11.25
Random UUID V4 generation	281.01	1.03	1.84
Random UUID V7 generation	302.97	1.10	1.96

Note: Error denotes the 99.9% confidence-interval margin. Statistics are calculated over measurement iterations remaining after upper outlier removal.

3.2 Performance Comparison with UUID

Under normal load, Secure SKEID generation is approximately 1.71 times faster than UUID V7 (Table 7). Secure SKEID generation incurred approximately 17.35 ns of additional end-to-end operation cost relative to plaintext SKEID generation.

SKEID generation is approximately 1.85 times faster than UUID V7 despite embedding additional metadata (entity type, epoch) and computing a BLAKE3 keyed MAC within the same 128-bit footprint. Both plaintext and Secure SKEID generation report zero managed allocation per operation. Implementations in other managed runtimes may exhibit different allocation profiles.

SKID generation is approximately 35.39 times faster than UUID V7 and 33.35 times faster than UUID V4. This performance advantage stems from the deterministic bit-packing approach. SKID generation requires only an atomic counter increment, a cached timestamp read, and bitwise operations, with no cryptographic random number generation.

3.3 Throughput Analysis

The sequence capacity and aggregate sequence capacity bound theoretical generation throughput. The saturation measurements in Table 8 capture generation costs including waits at tick boundaries. Because the delay between batches is excluded, their reciprocals do not represent continuous sustained throughput.

SKEID and Secure SKEID generation include SKID generation and therefore incur the same sequence backpressure. The generation speed advantage over UUID V4 and V7 measured under normal load does not extend to saturation. Parsing and tier conversion with a provided identifier do not consume sequence capacity. Their saturation-run measurements are provided in `supplementary-saturation-report.html`. All measured Source-Known operations reported zero managed allocation per operation. These results support the architectural separation between identifier generation and identifier processing.

The sequence limits and topology constants underlying the capacity definitions are checked by the `PaperThroughputAnalysisTests` unit test suite in the reference implementation.

3.4 Storage Comparison

Table 1 compares primary key and external identifier sizes across schemes.

At 8 bytes per SKID versus 16 bytes for UUID, raw key storage requirements are halved. Compared to KSUIDs at 20 bytes, raw key storage is reduced by a factor of 2.50 per primary key. Smaller keys can increase the number of keys per B-tree page and improve cache utilization. Effects on page splitting depend on the workload and index configuration. The timestamp-leading layout favors append-heavy workloads. Cursor-based pagination is natively supported through the SKID's numeric ordering (`WHERE id > :lastId ORDER BY id LIMIT :pageSize`), eliminating offset-based pagination overhead. Compared with the dual-identifier baseline, SKIDs reduce raw identifier column storage from 24 to 8 bytes per record (66.67%), deriving the external representation on demand.

Tier transformations had sub-microsecond mean execution times in the benchmark environment (Table 7), with no database or network access. Assuming millisecond-scale latency for I/O-bound identifier resolution, the measured transformation cost is negligible by comparison.

801 64-bit SKIDs are timestamp-ordered signed integers within a configured epoch, and plaintext
802 SKEIDs preserve timestamp order under lexicographic comparison of their specified big-endian
803 representation. These properties support an expectation of index locality similar to conventional
804 time-ordered integer and UUID keys (Davis, Peabody & Leach, 2024, sec. 2.1 and 6.11). However,
805 database-engine benchmarks were not conducted, so improvements in index size, insertion throughput,
806 and query latency remain unverified.

807 **4 DISCUSSION**

808 **4.1 Interpretation of Results**

809 The analysis indicates that the six target properties can be distributed across identity representations
810 under the stated cryptographic assumptions, provided that deployment coordination preserves
811 uniqueness, epoch context is retained, and keys and acceptance policies are managed consistently. In the
812 benchmark environment, generation across all three tiers was faster on average than UUID V4 and V7
813 within sequence capacity, but slower under saturation, with mean generation times remaining below one
814 microsecond in both conditions.

815 The measurements distinguish computational cost within sequence capacity (Table 7) from
816 generation cost under backpressure (Table 8). SKID's deterministic generation avoids the cryptographic
817 random number generation required by the measured UUID implementations. Secure SKEID parsing
818 was the most expensive Source-Known operation under normal load, while generation costs across the
819 three tiers converged under saturation because of shared sequence backpressure.

820 Zero-allocation behavior in core SKID and SKEID operations is important for managed runtime
821 environments (e.g., .NET, JVM) where garbage collection pauses can impact tail latency. All measured
822 SKID, SKEID, and Secure SKEID operations report zero managed allocation per operation, including
823 secure generation, parsing, and tier conversion.

824 Expected improvements in database index locality, insertion throughput, query latency, and
825 garbage-collection-related tail latency remain untested. Whether a distributed application operates
826 below the aggregate sequence capacity likewise depends on its workload and available resources. When
827 a sequence is exhausted, backpressure delays further generation for the affected application/entity-type
828 pair on that instance until the next tick, preserving uniqueness within that scope.

829 **4.2 Implications for Distributed Applications**

830 A practical implication is identifier validation at an API gateway or other trusted edge component
831 provisioned with the required keys. Such a component can check identifier integrity and expected
832 application and entity-type metadata without querying the backing database, allowing invalid identifiers
833 to be rejected before downstream processing. These checks do not establish record existence or replace
834 authentication and authorization.

835 This separation is also useful in eventually consistent systems, where a receiving service can
836 validate an identifier before the corresponding record becomes visible in its local read model. Identifier
837 validation requires no record lookup, while handling replication lag and temporary record absence
838 remains an application responsibility.

839 These implications follow from the separation of identifier validation from record lookup. Their
840 effects on end-to-end latency and throughput in deployed systems were not evaluated.

841 **4.3 Comparison with Existing Work**

842 Compared to UUID V7 (Davis, Peabody & Leach, 2024), the most directly comparable recent standard,
843 the three-tier identity system offers origin metadata, integrity verification (MAC), and confidentiality
844 (encryption) that UUID V7 lacks. UUID V7 provides millisecond-precision timestamps while SKID
845 uses 250-millisecond tick precision (see Design Trade-offs for the rationale and throughput analysis of
846 this choice).

847 Snowflake (Twitter Engineering, 2012) shares the same compact 64-bit footprint as SKID, but
848 SKEID adds integrity verification and Secure SKEID adds confidentiality, neither of which Snowflake
849 provides. Snowflake's 10-bit machine ID (1,024 machines) is comparable to SKID's combined 13-bit
850 topology (7-bit app ID + 6-bit app instance ID), but SKEID adds explicit entity type discrimination.
851 Both require coordination to assign machine/instance identifiers and both schemes converge on similar

single-epoch durations. However, Snowflake’s epoch is non-extensible while SKEID provides extended epoch coverage. Lastly, Snowflake uses 41-bit millisecond timestamps while SKID achieves comparable coverage with a 32-bit 250-millisecond timestamp and an epoch-half bit.

ULIDs (Feerasta, 2019) require twice the primary key storage (16 bytes versus 8 for SKID) and carry less embedded metadata. ULIDs use a 48-bit millisecond timestamp with ~8,919 years of coverage from the Unix epoch, while SKEIDs provide longer coverage through epoch extension. ULIDs rely on an 80-bit randomness field for probabilistic uniqueness across independent generators, while SKIDs use deterministic topology-based partitioning. ULIDs use a 26-character Crockford Base32 encoding that sorts lexicographically, targeting string-first environments, but lack UUID format compatibility. SKEIDs are valid RFC 9562 Version 8 UUIDs. ULIDs provide neither integrity verification nor confidentiality.

KSUIDs (Segment, 2023) use 20 bytes per binary identifier (see Table 1). Source-Known Identifiers provide more compact representations (8/16 bytes) with richer metadata. KSUIDs use second-precision timestamps while SKIDs use 250-millisecond ticks. KSUIDs provide ~136 years coverage with 32-bit seconds and no extension mechanism, whereas SKEIDs support epoch extension. KSUID’s 128-bit random payload trades storage efficiency for stronger per-identifier collision resistance without coordination. Source-Known Identifiers achieve uniqueness through the topology-based partitioning scheme with deployment time coordination, which guarantees uniqueness by construction rather than by statistical improbability. Additionally, the three-tier identity system provides entity type discrimination, integrity verification, and confidentiality, none of which KSUID offers. Even at the external-facing tier, Secure SKEIDs (16 bytes) remain more compact than KSUIDs (20 bytes), while the three-tier identity system satisfies all six desired identifier properties. KSUIDs use a custom 160-bit format with binary and Base62 representations and are not compatible with standard UUID parsers.

The original CUID (Elliott, 2023) was deprecated by its maintainer over security concerns associated with its timestamp-prefixed structure, as discussed in the Related Work section. Secure SKEIDs address this metadata exposure directly. The AES-256 encryption layer encrypts the entire SKEID payload, concealing its timestamp and source metadata from external consumers without the key.

CUID2 provides probabilistic uniqueness, whereas the three-tier identity system provides deterministic uniqueness within configured deployment partitions. For time-based sorting, CUID2 recommends a separate indexed `createdAt` column (Elliott, 2026). Source-Known Identifiers can avoid that additional column when their embedded generation timestamp satisfies the application’s requirements.

Tables 9 and 10 compare the security posture of the three-tier identity system against standard and alternative identifier schemes.

Table 9: Security comparison of standard identifier schemes.

Threat	SKEID / Secure SKEID	UUID V4	UUID V7
Structure-based enumeration	YES / NO (AES-256)	122-bit random	Timestamp + random/monotonic fields
Forgery	MAC verification	No detection	No detection
Embedded metadata leakage	YES / NO (AES-256)	None (random)	Timestamp visible
Cross-type confusion	Entity type check	Possible	Possible

Table 10: Security comparison of alternative identifier schemes.

Threat	SKEID / Secure SKEID	ULID	CUID2	KSUID	Snowflake
Structure-based enumeration	YES / NO (AES-256)	TS + 80-bit random	Hash-based	TS + 128-bit random	Timestamp-ordered
Forgery	MAC verification	No detection	No detection	No detection	No detection
Embedded metadata leakage	YES / NO (AES-256)	Timestamp visible	None (hashed)	Timestamp visible	TS + worker visible
Cross-type confusion	Entity type check	Possible	Possible	Possible	Possible

Note: TS denotes timestamp.

890 4.4 Threat Analysis

891 To evaluate the security posture of the three-tier identity system systematically, we apply a threat
892 analysis informed by the STRIDE categories (Shostack, 2014) (Spoofing, Tampering, Repudiation,
893 Information Disclosure, Denial of Service, Elevation of Privilege) to the three-tier identity model.

894 **Spoofing (Identifier Fabrication):** An attacker who intercepts a Secure SKEID cannot feasibly
895 derive the plaintext SKEID without the AES-256 key under the PRP assumption. Forging a valid Secure
896 SKEID requires producing ciphertext that, when decrypted, passes the applicable marker, MAC, epoch,
897 and expected-identity checks. Random-ciphertext acceptance probabilities are quantified above under
898 the conditions stated for Table 6. For plaintext SKEIDs within trusted environments, resistance to
899 identifier fabrication is limited by the 32-bit MAC and the number of verification attempts.

900 **Tampering (Identifier Modification):** Any modification to a Secure SKEID produces different
901 plaintext upon decryption under the same AES key, which must still pass the applicable validation
902 checks. For plaintext SKEIDs, the MAC authenticates the encoded fields, but a guessed tag for modified
903 fields can still verify with probability approximately 2^{-32} per attempt under the keyed-MAC assumption.

904 **Repudiation:** The three-tier identity system does not provide non-repudiation in the cryptographic
905 sense because it does not use digital signatures. However, the embedded topology metadata (app ID,
906 app instance ID, entity type) and timestamp provide forensic traceability. For a given SKEID, the
907 encoded application, instance, entity type, and approximate creation time can be recovered.

908 **Information Disclosure:** Plaintext SKEIDs deliberately expose metadata (timestamp, entity type,
909 topology) within trusted environments where this information supports routing and validation. For
910 external consumers, Secure SKEIDs encrypt the entire identifier using AES-256-ECB, which functions
911 as a pseudorandom permutation on the single 128-bit block. Under the PRP assumption, encryption
912 conceals the encoded timestamp, entity type, and topology from consumers without the key. Repeated
913 encryption of the same SKEID plaintext under the same key produces the same ciphertext, allowing
914 repeated observations of an identifier to be linked.

915 **Denial of Service:** Exhausting the sequence capacity triggers backpressure for the affected
916 application/entity-type pair on that instance. Its effect on system-wide availability depends on request
917 routing and shared resource limits. Parse operations (both encrypted and plaintext) are bounded-time
918 with no external dependencies, preventing parse-amplification attacks that require database or network
919 lookups.

920 **Elevation of Privilege (Cross-Entity Confusion):** The 8-bit entity type discriminator enables
921 applications to reject an identifier for one entity type (e.g., User with entity type 1) submitted as another
922 (e.g., AdminRole with entity type 5). The unchanged identifier still passes MAC verification but fails
923 the expected-type comparison. Applications must separately authorize access to the referenced record,
924 even when the identifier is valid and matches the expected type (MITRE Corporation, 2026a).

925 4.5 Design Trade-offs

926 **32-bit truncated MAC versus full MAC:** The 4-byte MAC is shorter than typical MACs (16–32 bytes)
927 but is one layer in a defense-in-depth strategy. The MAC is not the sole security mechanism. It works in
928 concert with AES encryption, marker detection, entity type matching, and record existence probability.

929 **250-millisecond tick versus millisecond precision:** 64-bit SKIDs use 250-millisecond tick
930 timestamps (32 bits, four ticks per second) instead of millisecond-precision (48 bits in UUID V7). This
931 sub-second quantization provides finer ordering than second-precision schemes while preserving
932 sufficient bits for application topology and sequence fields within 64 bits. Applications requiring finer
933 timestamp precision would need a different allocation of bits or a separate timestamp. Alternative SKID
934 profiles remain future work.

935 **Clock synchronization versus drift protection:** The SKID generator delegates wall-clock
936 synchronization to the operating system. Accurate timekeeping through Network Time Protocol (NTP)
937 (Mills et al., 2010), Precision Time Protocol (PTP) (IEEE, 2020), or equivalent protocols is an
938 infrastructure responsibility outside the scope of the identifier generator. The clock drift protection
939 mechanism described in the SKID specification handles only the consequences of clock jumps or drifts.
940 External coordination is required for app instance ID assignment as discussed in Limitation 2.
941 Timestamp monotonicity is guaranteed only within a running process.

Coordination versus probabilistic uniqueness: The three-tier identity system requires deployment-time coordination to assign app ID (7 bits, up to 128 applications) and app instance ID (6 bits, up to 64 instances per application). This stands in contrast to UUID V4, UUID V7, and CUID2, which achieve uniqueness without any coordination through cryptographic random number generation. The coordination cost is an intentional architectural trade-off. Deterministic uniqueness by construction eliminates the residual collision probability inherent in probabilistic schemes and allows the compact 64-bit representation that probabilistic schemes cannot achieve at equivalent uniqueness guarantees.

Topology field sizing: The 7-bit application field and 6-bit instance field bound the default profile to 128 applications and 64 instance identifiers per application to keep coordination manageable. The potential number of pairwise communication relationships $(n(n-1)/2)$ motivates this conservative design assumption by analogy with Brooks's discussion of communication overhead (Brooks, 1995, chap. 2). For 128 applications, this yields 8,128 possible paths. Well-designed systems mitigate this coordination complexity through bounded contexts (Evans, 2003) rather than unbounded application proliferation. Similarly, we assume that beyond approximately 10 concurrently active instances per application, resource contention on shared infrastructure (databases, message brokers, etc.) becomes the throughput bottleneck before the identifier generator itself. The 6-bit field accommodates up to 64 instance identifiers, providing headroom for operational needs such as rolling deployments and application restarts triggered by clock drift protection, where the restarting instance must acquire a new instance identifier to prevent duplicate identifiers. These are design limits, not empirically established thresholds for application scale or infrastructure contention. Domain-tailored bit layouts (see Future Work, item 4) remain available for specialized deployments requiring different trade-off profiles.

4.6 Limitations

1. **Uniqueness scope:** 64-bit SKIDs are unique per entity type within a deployment and epoch. This requires correct topology assignment, clock handling, sequence management, and safe instance reuse. Different entity types may share the same SKID value.
2. **Topology limits and coordination requirement:** The default profile supports at most 8,192 distinct generators (128 applications $\times$ 64 instances). Topology assignment (app ID and app instance ID) requires external coordination, unlike UUID V4/V7 which achieve uniqueness without coordination. The design assumes that a configuration or coordination service assigns a different app instance ID on every application restart, including ordinary restarts, and manages rotation through the 64-value pool. The identifier generator does not implement assignment or safe reuse. An app instance ID must not be assigned concurrently to multiple processes within the same application. Reuse within the same epoch requires the previous holder to have stopped and the new generator's timestamp to exceed the durably recorded maximum timestamp previously issued under that app ID/app instance ID pair. If no identifier can be safely assigned, startup must wait or fail.
3. **Key dependency:** SKEIDs and Secure SKEIDs require key material for generation and parsing. Loss of key material makes existing SKEIDs unparseable, though the underlying 64-bit SKIDs in the database remain valid and recoverable.
4. **Security scope:** The security analysis uses STRIDE-based threat modeling and quantitative probability analysis within the defense-in-depth architecture. It does not provide a formal security proof for the composition of AES encryption, the truncated BLAKE3 MAC, and the collision guard mechanism. The confidentiality analysis relies on modeling AES-256 as a secure pseudorandom permutation. Repeated encryption of the same SKEID plaintext under the same key produces the same ciphertext, allowing repeated observations of an identifier to be linked.
5. **Epoch-scoped storage efficiency:** The 8-byte SKID storage efficiency claim applies within a single epoch (~68 years). The 64-bit SKID encodes the time within the configured epoch but does not carry the epoch index itself. A database spanning multiple epochs has the following options.
 1. A composite key storing the epoch byte separately alongside the SKID. This adds per-row overhead that negates the 8-byte claim.

- 992 2. Migration to the 16-byte SKEID, which includes the epoch at byte 0. This also negates the
993 8-byte claim.
- 994 3. Epoch-partitioned storage, where the table or archive name implicitly carries the epoch
995 context (e.g., Entity_Epoch0, Entity_Epoch1). This preserves 8-byte rows by encoding
996 the epoch in the partition structure rather than in each record.

997 The 8-byte storage representation is suitable when the dataset remains within one epoch or epoch
998 context is maintained through partitioning. Long-lived archives and datasets spanning epochs require an
999 explicit storage and migration strategy. The addressable epoch range alone does not provide that
1000 strategy.

- 1001 6. **Secure SKEID UUID format compliance:** A Secure SKEID matches the UUID length and
1002 string format (32 hexadecimal digits in 8-4-4-4-12 grouping) and is accepted by PostgreSQL
1003 uuid columns and .NET Guid data-type parsers. This compatibility is validated by the reference
1004 implementation's UuidVersionAndVariantBatchCompatibilityTests,
1005 SourceKnownEntityIdPostgresUuidCompatibilityTests and
1006 SourceKnownEntityIdJsonCompatibilityTests integration tests (Kılıç, 2026c). However,
1007 AES-256 encryption replaces the plaintext Version and Variant marker bytes with pseudorandom
1008 ciphertext. The encrypted output is not guaranteed to preserve the version and variant bits
1009 required by RFC 9562 Sections 4.1 and 5.8. Strict RFC 9562 validators may therefore reject a
1010 Secure SKEID. Systems requiring strict RFC 9562 compliance at the validator level should
1011 accept Secure SKEIDs as opaque strings or 128-bit integers.

1012 4.7 Future Work

1013 Future work concerns five areas where the present analysis and implementation leave open questions.

- 1014 1. **Compositional analysis:** A formal analysis is needed to establish which security guarantees
1015 follow from combining single-block AES encryption, the truncated BLAKE3 MAC, and
1016 collision-guard verification under explicit assumptions. The present threat model and probability
1017 estimates do not provide such a proof.
- 1018 2. **AES-CMAC evaluation:** Integrity verification was initially planned with HMAC-SHA-256, then
1019 shifted toward SHA-3 motivated by potential post-quantum security benefits, and subsequently to
1020 BLAKE3 (descended from the SHA-3 finalist BLAKE) after benchmarks confirmed its speed
1021 advantage on small inputs. However, we discovered that single-block hardware-accelerated AES
1022 with cached keys may offer better performance. We plan to evaluate AES-256-CMAC as specified
1023 by the CMAC construction in NIST SP 800-38B (Dworkin, 2005). AES-256 retains a 128-bit
1024 block size, matching SKEID's 16-byte authentication input. We plan to retain BLAKE3's
1025 domain-separated key derivation mode (derive_key) for generating independent encryption and
1026 authentication keys from master key material in the reference implementation.
- 1027 3. **Operational lifecycle:** The mechanism for transitioning between epochs is not specified. Given
1028 that epoch 0x00 spans until approximately 2093 CE, this is left as future work. Future work must
1029 address how stored identifiers retain their epoch context during migration and across long-lived
1030 datasets. Key-ring fallback is configurable in the reference implementation, but coordinated key
1031 rollout, revocation, and cross-key ambiguity as the verification key set changes remain application
1032 responsibilities and subjects for further study. Instance identifier assignment and safe reuse across
1033 restarts likewise require an external coordination mechanism, as described in Limitation 2.
- 1034 4. **Portability and interoperability:** An author-hosted draft specification (draft-kilic-skid-00)
1035 in Internet-Draft format (Kılıç, 2026b) is available in the archived DRN-Project v0.10.0 release,
1036 containing algorithms, CDDL definitions, and test vectors. Formal IETF submission remains a
1037 separate dissemination step. Independent implementations in other languages would test the
1038 draft's interoperability through shared test vectors. Alternative bit allocations would require
1039 explicit trade-offs among timestamp precision, epoch duration, topology capacity, and sequence
1040 capacity, together with compatibility rules and validation.

1041 **5. Platform-specific UUID compatibility for Secure SKEID:** The PostgreSQL and .NET
1042 integration tests demonstrate that Secure SKEIDs can be stored and parsed without preserving
1043 UUIDv8 version and variant bits. RFC 9562 recommends treating UUIDs opaquely and avoiding
1044 unnecessary inspection (Davis, Peabody & Leach, 2024, sec. 6.12). It also defines Nil and Max
1045 UUIDs with all 128 bits set to zero or one, respectively, and version values 9–15 remain reserved
1046 for future definitions rather than being permanently excluded from the UUID version space
1047 (Davis, Peabody & Leach, 2024, secs. 4.2, 5.9, and 5.10). These provisions distinguish general
1048 UUID handling from validation of a particular version, but do not establish RFC conformance for
1049 arbitrary ciphertext or require parsers to accept it. The current results suggest that preserving
1050 UUIDv8 bits may also be unnecessary on other contemporary platforms. Whether this holds for
1051 most practical deployments remains to be established. Documenting compatibility across
1052 platform and library versions is left as future work. Where the complete application path accepts
1053 Secure SKEIDs, techniques such as cycle walking or specialized Feistel constructions (Khuong,
1054 2022) may be unnecessary solely for UUID compatibility.

1055 **5 CONCLUSIONS**

1056 Modern distributed systems face competing demands for storage efficiency, chronological ordering, and
1057 external confidentiality. Dual-identifier schemas try to address chronological ordering and external
1058 confidentiality at the cost of additional operational and storage overhead. This paper demonstrates that
1059 these goals are not fundamentally incompatible at the system level when a single entity identity is
1060 represented differently across trust boundaries.

1061 The proposed three-tier architecture aligns SKID (64-bit, database), SKEID (128-bit, trusted
1062 internal), and Secure SKEID (128-bit, external) with the trust boundaries commonly found in
1063 distributed applications. Deterministic bidirectional transformations enable direct conversion between
1064 tiers. The defense-in-depth security architecture combines confidentiality, identifier authentication, and
1065 application-level validation, with random-ciphertext acceptance probabilities depending on the enabled
1066 checks.

1067 In the benchmark under normal load within sequence capacity, generation of SKIDs, SKEIDs, and
1068 Secure SKEIDs is faster than generation of either UUID V4 or UUID V7. Under saturation,
1069 backpressure raises Source-Known generation means above the UUID baselines. All measured
1070 identifier generation, parsing, and conversion operations in both runs have sub-microsecond mean
1071 execution times.

1072 The open-source reference implementation, published as NuGet packages with integration and unit
1073 test coverage, is available for community review and adoption. A draft specification prepared in
1074 Internet-Draft format is available (Kılıç, 2026b) to support cross-platform implementation and
1075 interoperability testing. Formal submission to the IETF remains future work.

1076 Within the stated deployment and key-management constraints, the design illustrates how an entity's
1077 identity can remain stable while its representation changes across trust boundaries to satisfy all six
1078 target properties.

1079 **DATA AVAILABILITY**

1080 The source code for DRN-Project is available on GitHub. Version v0.10.0, released on 20 September
1081 2026, is archived on Zenodo (DOI: 10.5281/zenodo.22863243) and available under the GitHub tag
1082 release/v0.10.0. Benchmark data (CSV) and rendered reports (HTML) are included as six
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1090 Duran Serkan Kılıç conducted the research, conceived the design, implemented the software, conducted
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